

the average user, computers displayed or printed colour images where the colour balance was obviously wrong. In the early days of colour computing this is frequently exactly what did happen.

This problem has been solved by building automatic colour management into the operating system and into printer drivers. It works by using a colour space that has a relatively small gamut for reference and by assuming that, by default, everything in the colour work flow conforms to that colour space. The space used, sRGB (standard Red Green Blue) was originally adopted by Microsoft and HP for standardising colour on web pages and is based on the colour space of the typical computer monitor (at the time, this was CRT). Therefore most monitors conform reasonably well to sRGB without any calibration or profile. Many digital cameras, especially the consumer models, use sRGB as their default colour space, although it may be possible to select other colour spaces.

Instead of individual colour profiles measured from the actual devices being used, manufacturers' generic profiles are used. Generic profiles, which allow for a certain amount of manufacturing variation, are usually included with most colour devices, or are available from websites.

When it comes to printing, all printer manufacturers have a selection of presets included in their printer drivers designed for a range of papers, although these are of course always designed for use with the printer manufacturer's own brand of papers. These profiles are often supplemented by settings for things like "best quality photo" or "text and image" which are effectively simple ways of setting the colour intent.

Although automatic sRGB works reasonably well and is a lot cheaper than a fully colour managed system, intentional colour management allows more accurate control of colour and can provide better results.

The Colour Management Chain – Open Or Closed Loop

A colour management work flow can be regarded as a closed loop when the characteristics of the final reproduction device can be measured and fed back into the work flow. If the final colour output is to be a commercial printing press the ideal would be to obtain an individual device profile for the press, inks and paper used and to use that to create the output file. With such a profile it is possible to close the management loop. Commercial print press operators in general have always proved to be a huge stumbling block in this respect. It is normally extremely difficult to establish a meaningful dialogue with commercial print-

ers on the subject of colour in general or on press profiling.

Most printers will supply hard copy colour proofs on request and for a fee. The problem with these is that usually they are not produced using the same press characteristics, inks and paper used for the final print run, so they are only useful for showing really gross errors.

What is possible is to establish to which standards a commercial printer says they calibrate their presses, and what type of paper they will use. This is likely to be one of the commercial standard profiles which are supplied with leading image editing software. Adobe Photoshop for example has profiles for SWOP (Standard Web Offset Press), Euroscale coated V2 and so on.

In Conclusion

In essence, colour management simply consist in choosing and applying the correct colour spaces and providing accurate device profiles at every step of the colour management work flow. In the past, poor software implementation, lack of accurate low cost hardware and software tools and poor support for colour management within the operating system, have made it hard for the average person to establish a colour management work flow that actually works.

Fortunately over the years the situation has improved and software and hardware colour management systems that leverage the latest advances in technology, such as high intensity LED's, instead of exotic gas lamps, are becoming available. It is now possible to set up a calibrated and profiled colour work flow on an ordinary desktop PC at reasonable cost. ☑



Resources And References

<http://www.microsoft.com/downloads/details.aspx?FamilyID=1e33dca0-7721-43ca-9174-7f8d429fbb9e&displaylang=en> Microsoft Color Control Panel Applet for Windows XP

www.normankoren.com Photographer Norman Koren's website with information on editing digital photographs and on colour management.

www.poynton.com - Charles Poynton's essays on various colour related subjects.

http://en.wikipedia.org/wiki/Lab_color_space - Wikipedia article on L*a*b* colour space.

<http://en.wikipedia.org/wiki/sRGB> - Wikipedia page on the sRGB colour space.

www.color.org - The International Color Consortium site. Lots of information on colour management and particularly on ICC profiles.

<http://www.brucelindbloom.com/index.html?WorkingSpaceInfo.html> - 3D versions of various colour spaces which can be viewed and rotated on Bruce Lindbloom's website.

Real World Colour Management - 2nd Ed. By Bruce Fraser, Chris Murphy & Fred Bunting, published by Peachpit Press ISBN 0-321-26722-2